A Two-Pronged Approach to Differential Gene Expression

High throughput sequencing technologies have generated *exabytes* (10^{18} bytes) of genomics data critical to improving human health. Of course, as this dataset grows, so too does the challenge of interpreting it, particularly with respect to gene expression. One of the biggest challenges in differential gene expression profiling is the noise and its tendency to obfuscate insights. Noise here originates from the randomness of cellular life, the inherent instability of mRNA molecules and the accidental influence of the observer on the observed. To overcome this noise, we'll apply a two-pronged approach by co-analyzing gene expression and chromatin state where the intersection of the analyses gives a clearer view of the underlying biology.

Eukaryotic gene expression is primarily regulated through modification of chromatin structure. Within chromatin, negatively charged DNA is wrapped around positively charged histone proteins like beads on a string in what are called nucleosomes. Tightly bound DNA is inaccessible and thus silent, while free DNA is available for transcription. Specific chemical modifications to histones (i.e. H3K9ac, H3k27ac) loosen their attraction to DNA, allowing transcription, while repressive modifications (i.e. H3K9me3, H3K27me3) recruit protein complexes to condense nucleosomes to silence them. This is known as the epigenome or "Histone Code", a molecular language typed onto nucleosomes that regulate gene expression.

Recently, a group from MIT generated hundreds of reference epigenomes by assigning a "chromatin state" to specific stretches of DNA within different tissues. Briefly, they input thousands of datasets stemming from 18 epigenetic assays to ChromHMM, a machine-learning model that assigns "chromatin states" to stretches of DNA based on its combination of histone modifications. These reference epigenomes are available for 33 different tissues in both healthy and diseased states and can be used to study epigenetic changes in the context of disease.

Here, we've developed a pipeline to co-analyze reference epigenomes with gene expression data to gain a clearer view of the transcriptional changes in cancer. The pipeline requires input gene expression data (available from Genomic Data Commons by the National Cancer Institute) and reference epigenomes (from EpiMap) for both healthy and cancerous cells. As proof of concept, we've applied this pipeline, creatively named DEG CrossRef, to hepatocellular carcinoma (see Githhub.com/matthall1797/ DEG_CrossRef_HCC) and found several classified biomarkers, prognosis indicators and therapeutic targets. Gene set enrichment analysis identified cellular signals of proliferation, liver dedifferentiation and immune evasion, signifying how these cells have transformed and providing key insights to aid in developing precise treatments.

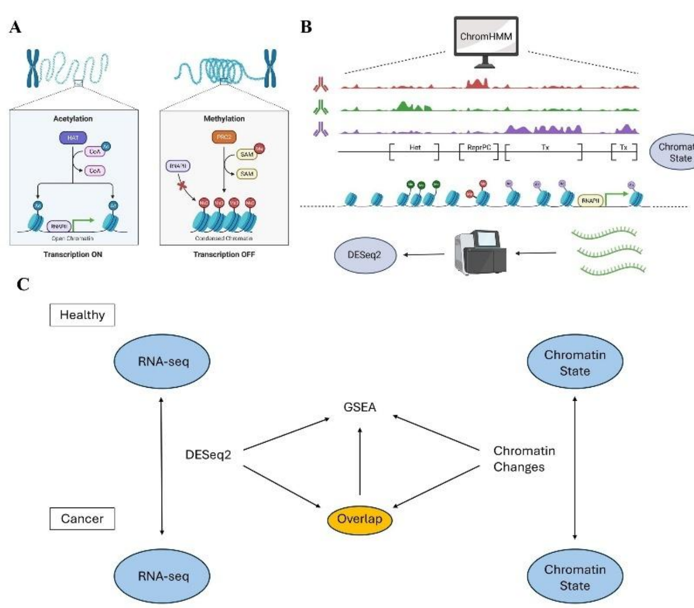

Figure 1. Graphical Overview of Pipeline. **A)** Diagram of Gene Regulation by Histone Modification. **B)** Comparison of the two Input Datasets for the Pipeline. Chromatin State is inferred by ChromHMM based on data from 18 Epigenetic Assays and DESeq2 is performed on RNA-seq datasets from the NCI. **C)** Graphical overview of the Pipeline.

Methods

Input Data: Required input data to the pipeline includes reference epigenomes and enhancer-gene linsk from EpiMap as well as Gene Expression Quantification from the GDC and the NCI, see the README file in https://github.com/matthall1797/DEG_CrossRef for requirements and instructions to attain compatible input data.

Pipeline: Run the pipeline through the Snakefile, using the command 'snakemake -s Snakefile -p --use-conda --cores -1' in a linux environment. The first time the pipeline is run, it will take ~20 mins to set up the required environments stated in the envs/ subdirectory. Within the pipeline, input data is first pre-processed to include sample names within .bed files and concatenated into a single file for healthy and cancer datasets. Identical features are merged into one. Enhancer-Gene links converted from hg17 to hg38 coordinates using liftOver. Enhancers are removed from the reference epigenomes ('chromatin_states') and Enhancer-Gene links are combined into a single reference epigenome for healthy and cancer datasets. Chromatin state changes are identified using bedtools intersect to find incongruent features that overlap between the healthy and cancer epigenomes, and false positives are removed where there is conflicting information: where a 'changed_state' overlaps a 'no_change' (i.e. where 'Tx-Het' overlaps 'Tx-Tx'). These overlapping features are aligned to transcripts and assigned a value: {'Tx-Het': -4, 'Tx-ReprPC': -3, 'Tx-ReprPCWk': -2, 'Tx-Quies': -1, 'Quies-Tx': 1, 'ReprPCWk-Tx': 2, 'ReprPC-Tx': 3, 'Het-Tx': 4}. DESeq2 analysis is completed in parallel using the Gene Expression Quantification data from the NCI (requiring the -01A and -11A suffixes and a Sample_Sheet.csv, see README). Finally genes with aligned changes in chromatin state and DESeq2 are extracted as the 'Overlap' dataset, and a 'combined_score' is calculated as the average of the log2FC (DESeq2) and chromatin_change value. GSEA is then performed independently on the Chromatin Change, DESeq2 and Overlap datasets.

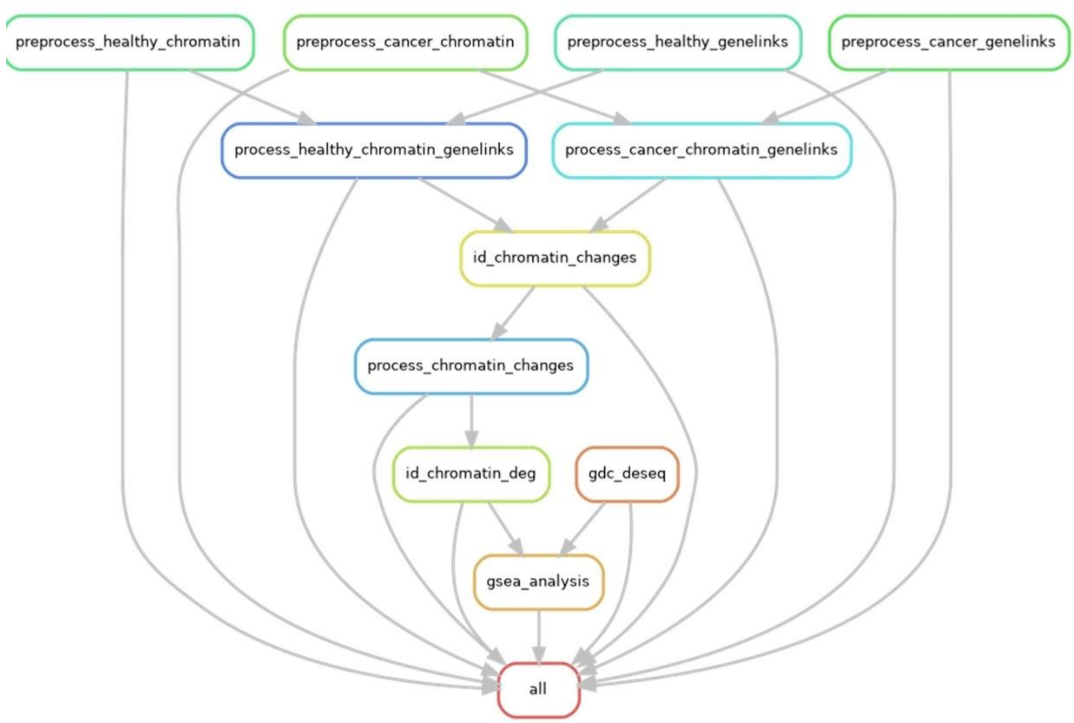

Figure 2. Directed Acyclic Graph for the DEG CrossRef Pipeline.

Results

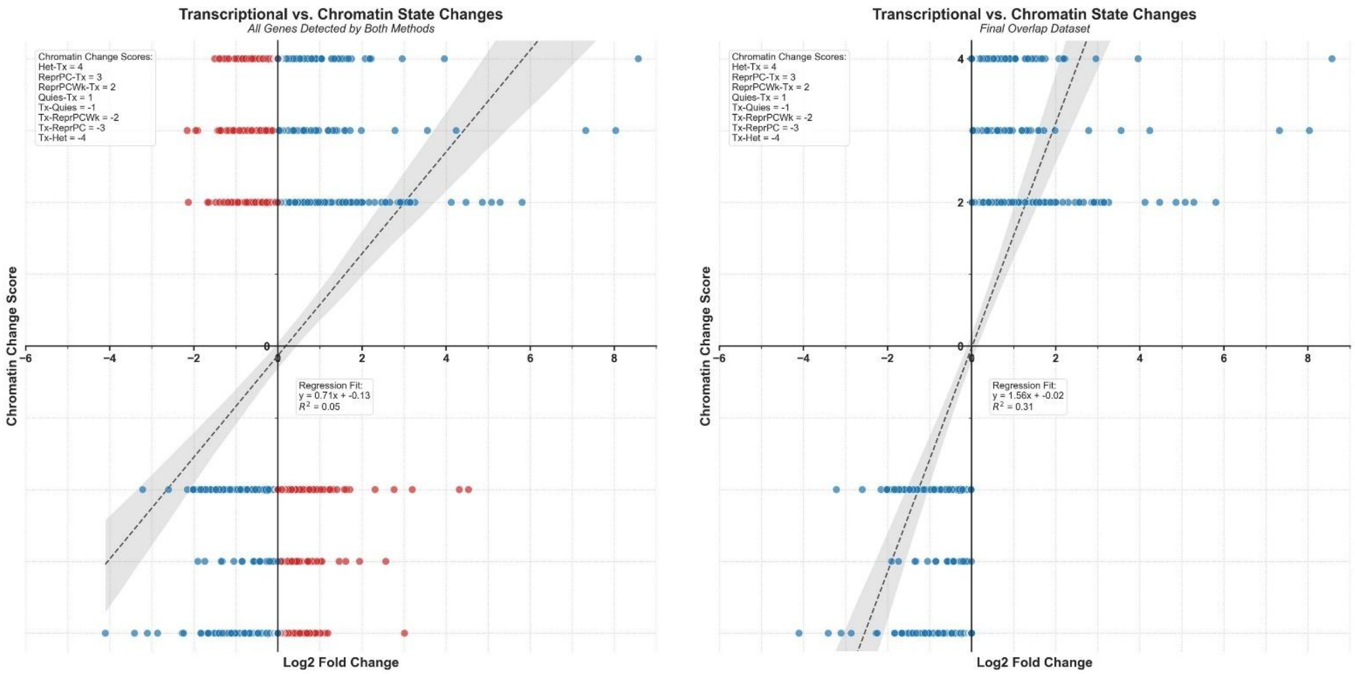

Figure 3. X-Y Plot comparing Log2 Fold Change with Chromatin Change Score for All Genes with Chromatin Changes **(Left)** and the final Overlap dataset with aligned DESeq2 and Chromatin Change changes **(Right)**.

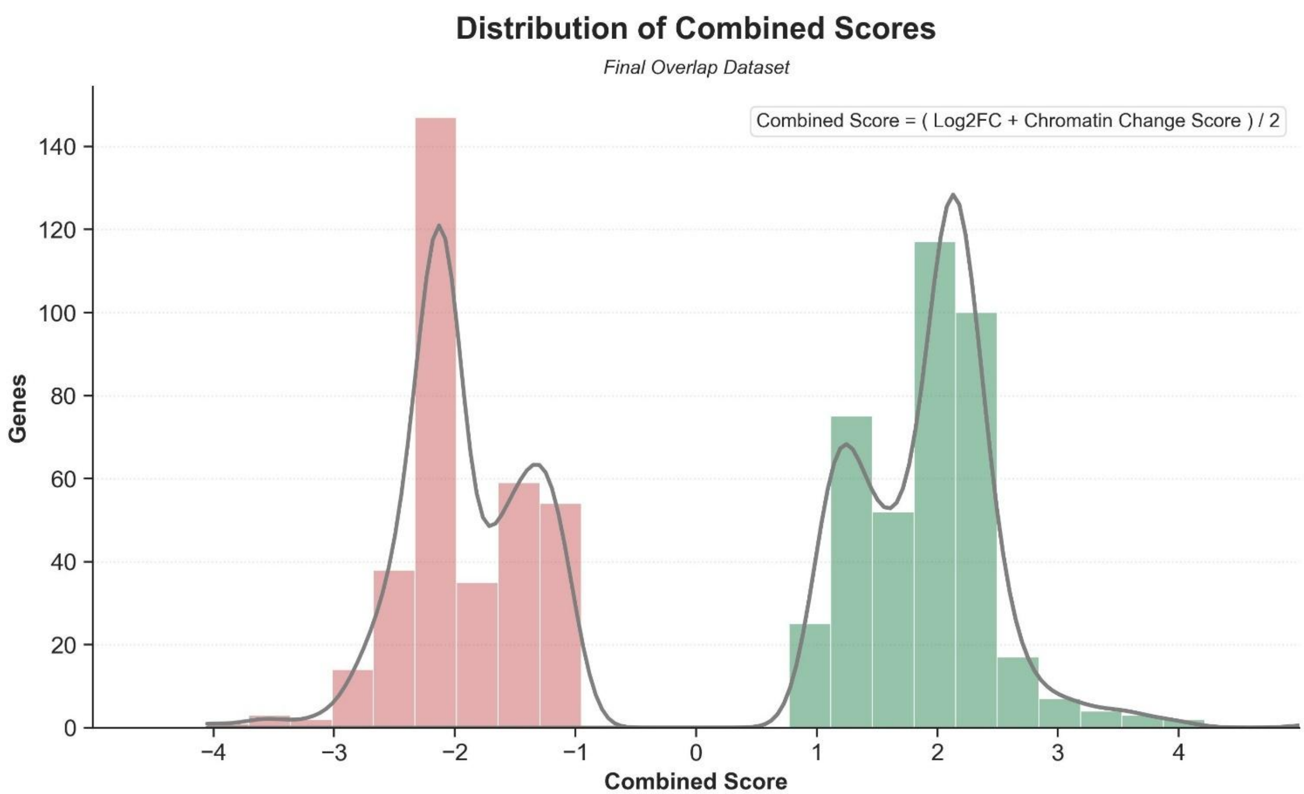

Figure 4. Distribution of Combined Score within the final Overlap Dataset.

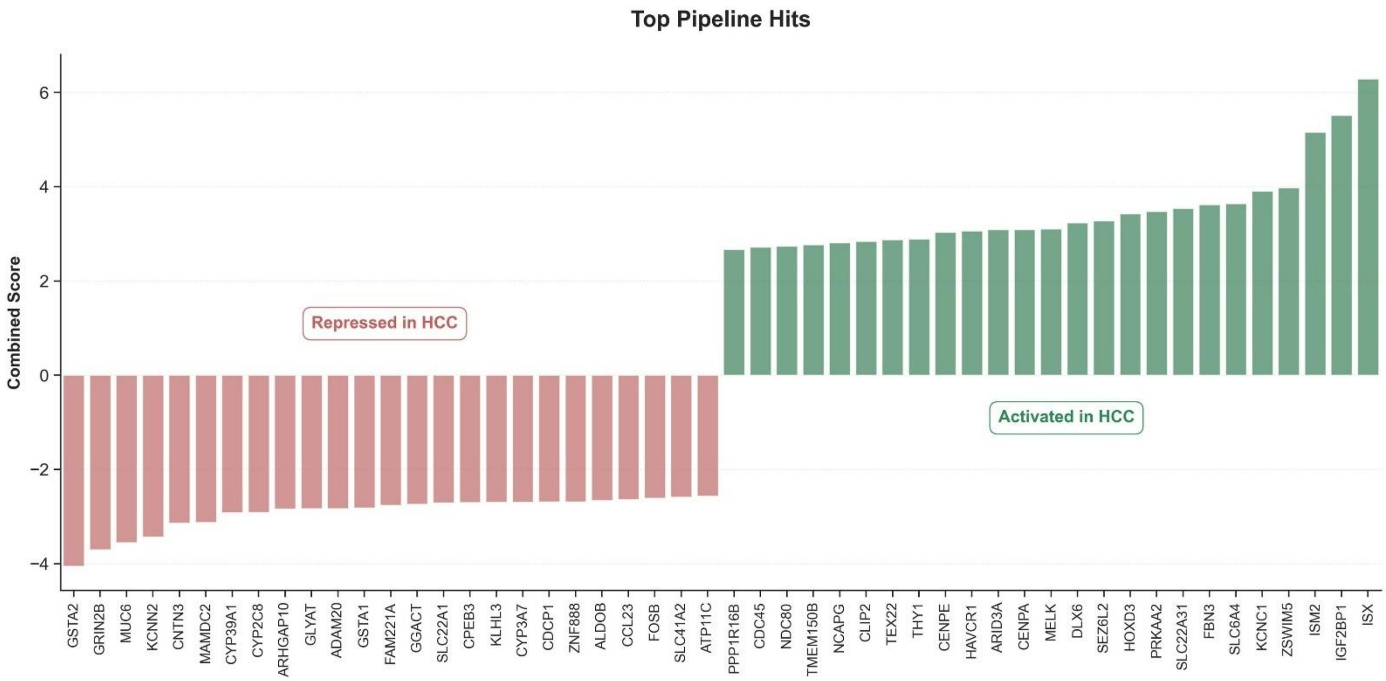

Figure 5. Top 25 Hits for both Activated and Repressed Gene Expression as determined by the DEG CrossRef Pipeline for Hepatocellular Carcinoma (HCC)

Gene	Log2FC	Chromatin Change	Combined Score	Notes
ISX*	8.571815	Het-Tx	6.285907	Master Regulator of Transformation, Immune Evasion, Poor Prognosis
IGF2BP1*	8.034589	ReprPC-Tx	5.517294	Stabilizes mRNA of Oncogenes, Inhibitors in Clinic
ISM2*	7.321957	ReprPC-Tx	5.160979	Immune Evasion, Poor Prognosis
ZSWIM5*	3.959452	Het-Tx	3.979726	Transcription Factor, EMT, Metastasis, Poor Prognosis
KCNC1	5.809389	ReprPCWk-Tx	3.904694	K+ Channel, Low baseMean
SLC6A4	5.284794	ReprPCWk-Tx	3.642397	Serotonin Transporter, Proliferation, Immune Evasion
FBN3*	4.238417	ReprPC-Tx	3.619209	Extracellular Matrix Protein, EMT, Metastasis
SLC22A31	5.081164	ReprPCWk-Tx	3.540582	Ion Transporter
PRKAA2*	2.957351	Het-Tx	3.478676	AMP-activated Protein Kinase, Immune Evasion, Metabolic Rewiring, Poor Prognosis
HOXD3	4.861216	ReprPCWk-Tx	3.430608	Transcription Factor, Low baseMean
SEZ6L2*	3.555353	ReprPC-Tx	3.277677	Cell Surface Protein, Expressed in Lung Cancers
DLX6	4.47294	ReprPCWk-Tx	3.23647	Antisense Transcript (DLX6-AS1) Oncogenic Biomarker/Target
MELK*	2.21963	Het-Tx	3.109815	Protein Kinase, Oncogene, Proliferation, Target of Clinical Trials
CENPA*	2.189096	Het-Tx	3.094548	H3 Variant in Centromeres, Genome Instability, Poor Prognosis
ARID3A	2.185149	Het-Tx	3.092574	Transcription Factor, Epigenetic Reprogramming, Dedifferentiation
HAVCR1	4.123269	ReprPCWk-Tx	3.061635	Transmembrane Glycoprotein, Immune Evasion, Proliferation, Poor Prognosis
CENPE*	2.075049	Het-Tx	3.037524	Kinesin, Chromosome Alignment, Genome Instability, Target in Clinical Trials
THY1*	2.784659	ReprPC-Tx	2.89233	Cell Surface Glycoprotein, Stemness, EMT, Metastasis, Poor Prognosis
TEX22	1.750108	Het-Tx	2.875054	Testis-specific, Low baseMean
CLIP2	1.678353	Het-Tx	2.839176	Microtubule Trafficking
NCAPG*	1.630599	Het-Tx	2.8153	Chromosome Condensation, Proliferation, Resistance to Ferroptosis, Poor Prognosis
TMEM150B	1.535851	Het-Tx	2.767926	Damage-Regulated Autophagy Modulator
NDC80*	1.488165	Het-Tx	2.744083	Kinetochore Component, Genome Instability, Poor Prognosis
CDC45	1.437945	Het-Tx	2.718973	DNA Replication Initiation, Proliferation, Interacts with IGF2BP2 , Poor Prognosis
PPP1R16B	1.343725	Het-Tx	2.671862	Regulator of Protein Phosphatase 1, Adhesion? Angiogenesis?

Table 1. Top 25 Upregulated Genes in HCC Transformation according to the Pipeline.

Gene	Log2FC	Chromatin Change	Combined Score	Notes
GSTA2*	-4.10546	Tx-Het	-4.05273	Oxidative Stress Response, Dedifferentiation
GRIN2B	-3.4106	Tx-Het	-3.7053	Neuronal Receptor, Low baseMean
MUC6*	-3.10457	Tx-Het	-3.55228	Dedifferentiation, EMT?
KCNN2	-2.86249	Tx-Het	-3.43125	Ca^{2+} Signaling
CNTN3	-2.27491	Tx-Het	-3.13746	Neuronal Cell-Adhesion, Cell Composition Effect?
MAMDC2*	-2.24438	Tx-Het	-3.12219	Tumor Suppressor, Poor Prognosis
CYP39A1	-1.83399	Tx-Het	-2.91699	Bile Acid Synthesis, Dedifferentiation
CYP2C8	-1.81841	Tx-Het	-2.90921	Fatty Acid/Drug Metabolism, Dedifferentiation
ARHGAP10*	-1.67166	Tx-Het	-2.83583	Rho GTPase–activating protein, Tumor Suppressor, EMT, Metastasis
GLYAT	-1.65571	Tx-Het	-2.82786	Acetyl-CoA Detox, Dedifferentiation
ADAM20	-1.65545	Tx-Het	-2.82773	Testis-Specific Adhesion, Low baseMean
GSTA1*	-1.6354	Tx-Het	-2.8177	Oxidative Stress Response, Dedifferentiation
FAM221A	-1.51211	Tx-Het	-2.75606	Uncharacterized, Cell Composition Effect?
GGACT	-1.47963	Tx-Het	-2.73982	Protein Turnover, Glutathione Metabolism, Tumor Progression
SLC22A1*	-1.41054	Tx-Het	-2.70527	Cation Transporter, Drug Importer, Resistance, Dedifferentiation
CPEB3*	-1.40236	Tx-Het	-2.70118	Tumor Suppressor, Proliferation
KLHL3	-1.39303	Tx-Het	-2.69651	Promotes Substrate Ubiquitination, Tolerance?
CYP3A7	-1.39053	Tx-Het	-2.69526	Drug Metabolism, Lipid Synthesis, Dedifferentiation?
CDCP1	-1.37015	Tx-Het	-2.68507	Pro-Metastatic Transmembrane Protein, Expected Upregulation
ZNF888*	-1.36975	Tx-Het	-2.68488	Transcription Factor, EMT?
ALDOB*	-1.32239	Tx-Het	-2.6612	Gluconeogenesis, Dedifferentiation
CCL23*	-1.26749	Tx-Het	-2.63374	Chemokine, Immune Evasion
FOSB*	-3.21706	Tx-ReprPCWk	-2.60853	AP-1 Cofactor, Dedifferentiation, EMT, Metastasis
SLC41A2	-1.18147	Tx-Het	-2.59073	Mg^{2+} Transporter, DNA Instability?
ATP11C	-1.12861	Tx-Het	-2.56431	Phospholipid-Translocating ATPase Component

Table 2. Top 25 Downregulated Genes in HCC Transformation according to the Pipeline.

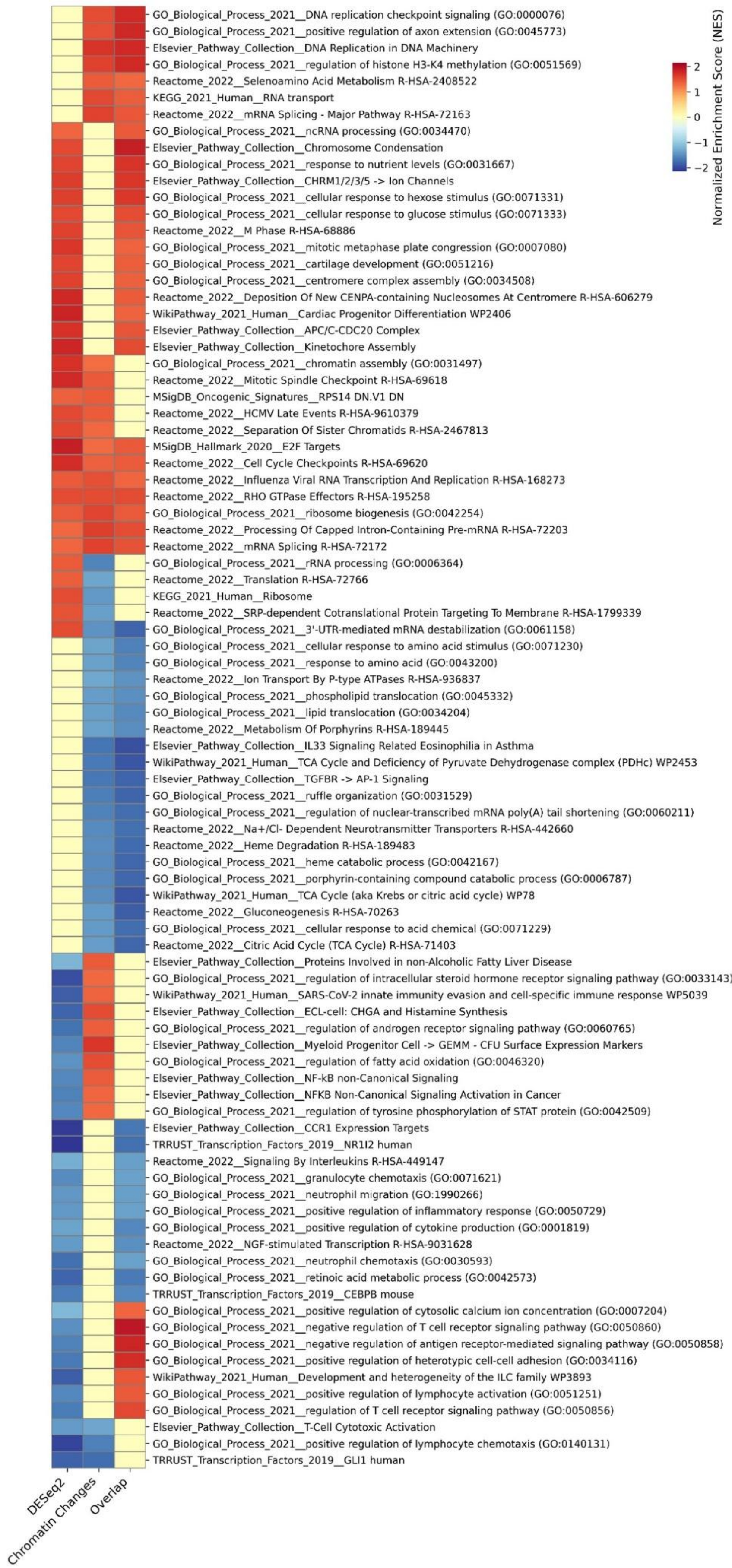

Figure 6. Gene Set Enrichment Analysis of the DESeq2, Chromatin Changes and Overlap Datasets

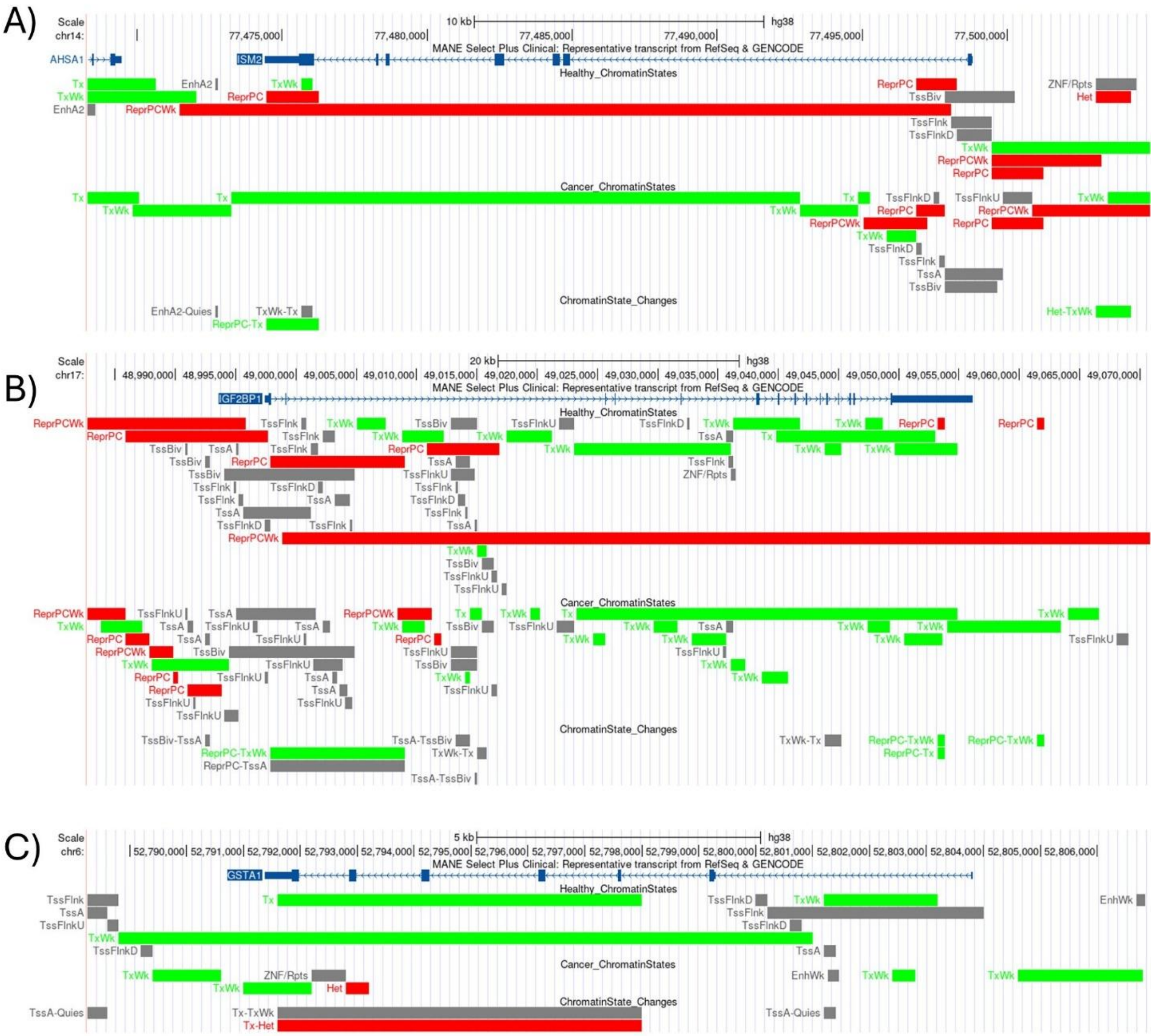

Figure 7. Representative Genome Browser Alignments for the Pipeline Hits **A)** ISM2 **B)** IGF2BP1 and **C)** GSTA1.